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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Computer Science & Engg.) (CBCS) Degree Examinations

May - June 2024

23CSE212 – DEEP LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions each from **Unit - I and Unit - II** and **One full** question from **Unit - III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define Deep Learning and explain its significance in modern AI applications.	10	L*2	1	1
b)	Outline five types of activation functions by plotting the graph for each.	10	L2	2	2
2. a)	Define Neural Network. With help of neat diagram, outline basics of Neural Networks.	10	L2	1	1
b)	Explain the concept of bias-variance tradeoff in the context of neural network models.	10	L2	2	2
3. a)	Explain various regularization learning strategies employed in deep learning.	10	L2	1	1
b)	What is Back propagation? Examine how learning rate used in back propagation network.	10	L2	2	2
Unit – II					
4. a)	Explain convolution and pooling operations in the context of Convolutional Neural Networks (CNNs).	10	L2	3	2
b)	Examine how scattering networks leverage wavelet transforms to extract features from signals?	10	L2	4	1
5. a)	Explain invariance, stability, and variability in the context of neural network models.	10	L2	3	2
b)	Examine the connections between scattering networks and dictionary learning, LISTA (Learned ISTA), and how they influence each other in signal processing tasks.	10	L2	4	1
6. a)	Explain the concepts of deformation model and stochastic model in CNN.	10	L2	3	2
b)	Analyze the working of localization regression model with example.	10	L2	4	1
Unit – III					
7. a)	Discuss the architecture of a basic RNN cell and how it processes sequential inputs to produce sequential outputs.	10	L2	5	2
b)	Discuss the role of the gates in controlling the flow of gradients during backpropagation, allowing GRU to capture long-term dependencies more effectively compared to traditional RNNs.	10	L2	5	2
8. a)	Explain different types of Recurrent Neural Network with examples.	10	L2	5	2
b)	Examine how Gated Recurrent Unit (GRU) and Long Short-Term Memory units (LSTM) address the vanishing gradient problem encountered by traditional Recurrent Neural Networks (RNNs), elucidating their unique mechanisms for mitigating this challenge.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
